COHEN-MACAULAYNESS OF MORPHISMS OF SCHEMES IS STABLE UNDER FLAT BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5911-5926 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5928-5978, the complete author proof (49 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Cohen-Macaulay morphisms, depth and flat local ring maps, and base change of morphisms; the transfer is retained.

Source reference. Original lemma algebra-lemma-base-change-flat-up-down: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5978–5978. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma varieties-lemma-CM-base-change: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5978–5978. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma varieties-lemma-locally-Noetherian-base-change: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5978–5978. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume that all the fibres X_y are locally Noetherian schemes. Let $Y' \to Y$ be locally of finite type. Let $f' : X' \to Y'$ be the base change of f. Let $x' \in X'$ be a point with image $x \in X$.*

1. *If f is Cohen-Macaulay at x, then $f' : X' \to Y'$ is Cohen-Macaulay at x'.*
2. *If f is flat at x and f' is Cohen-Macaulay at x', then f is Cohen-Macaulay at x.*
3. *If $Y' \to Y$ is flat at $f'(x')$ and f' is Cohen-Macaulay at x', then f is Cohen-Macaulay at x.*

Proof. Note that the assumption on $Y' \to Y$ implies that for $y' \in Y'$ mapping to $y \in Y$ the field extension $\kappa(y')/\kappa(y)$ is finitely generated. Hence also all the fibres $X'_{y'} = (X_y)_{\kappa(y')}$ are locally Noetherian, see Varieties, Lemma varieties-lemma-locally-Noetherian-base-change. Thus the lemma makes sense. Set $y' = f'(x')$ and

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$y = f(x)$. Hence we get the following commutative diagram of local rings

$$\begin{array}{ccc} \mathcal{O}_{X',x'} & \longleftarrow & \mathcal{O}_{X,x} \\ \uparrow & & \uparrow \\ \mathcal{O}_{Y',y'} & \longleftarrow & \mathcal{O}_{Y,y} \end{array}$$

where the upper left corner is a localization of the tensor product of the upper right and lower left corners over the lower right corner.

Assume f is Cohen-Macaulay at x. The flatness of $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ implies the flatness of $\mathcal{O}_{Y',y'} \to \mathcal{O}_{X',x'}$, see Algebra, Lemma algebra-lemma-base-change-flat-up-down. The fact that $\mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ is Cohen-Macaulay implies that $\mathcal{O}_{X',x'}/\mathfrak{m}_{y'}\mathcal{O}_{X',x'}$ is Cohen-Macaulay, see Varieties, Lemma varieties-lemma-CM-base-change. Hence we see that f' is Cohen-Macaulay at x'.

Assume f is flat at x and f' is Cohen-Macaulay at x'. The fact that $\mathcal{O}_{X',x'}/\mathfrak{m}_{y'}\mathcal{O}_{X',x'}$ is Cohen-Macaulay implies that $\mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ is Cohen-Macaulay, see Varieties, Lemma varieties-lemma-CM-base-change. Hence we see that f is Cohen-Macaulay at x.

Assume $Y' \to Y$ is flat at y' and f' is Cohen-Macaulay at x'. The flatness of $\mathcal{O}_{Y',y'} \to \mathcal{O}_{X',x'}$ and $\mathcal{O}_{Y,y} \to \mathcal{O}_{Y',y'}$ implies the flatness of $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$, see Algebra, Lemma algebra-lemma-base-change-flat-up-down. The fact that $\mathcal{O}_{X',x'}/\mathfrak{m}_{y'}\mathcal{O}_{X',x'}$ is Cohen-Macaulay implies that $\mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ is Cohen-Macaulay, see Varieties, Lemma varieties-lemma-CM-base-change. Hence we see that f is Cohen-Macaulay at x. $\square$